

Answer Key — Biology Grade 10

Sub-Strand 2.1: Plant Nutrition • Lesson 2: Leaf Structure: The Architecture of a Photosynthetic Machine

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
D	C	D	A	D	C	B	C	B	A

USE AFTER: OBSERVE PHASE — Observe, Step 4: locate stomata on the lower epidermis and sketch a single stoma with its two guard cells

Q1. You are looking at a cross-section of a mango leaf. On which layer would you find most of the stomata, with a pair of guard cells around each one?

- A. The palisade mesophyll
- B. The waxy cuticle on the upper surface
- C. The xylem in the vascular bundle
- D. The lower epidermis**

Correct answer: D. Most stomata are on the lower epidermis, where each pore is surrounded by two guard cells. The palisade layer and xylem are inside the leaf, and the cuticle is only a waxy covering.

USE AFTER: OBSERVE PHASE — Observe, Step 3: shade and label every distinct layer on the printed outline diagram

Q2. In a leaf cross-section, which layer sits just below the upper epidermis and is made of tightly packed, column-shaped cells full of chloroplasts?

- A. Lower epidermis
- B. Spongy mesophyll
- C. Palisade mesophyll**
- D. Phloem

Correct answer: C. Palisade mesophyll cells are tightly packed near the upper surface and hold many chloroplasts, so they capture the most light. Spongy mesophyll lies below with loosely arranged cells and air spaces.

USE AFTER: EXPLAIN PHASE — Explain: expert group for spongy mesophyll/air spaces presents its function statement

Q3. The spongy mesophyll has many air spaces between its cells. What is the main advantage of these air spaces?

- A. They carry sugars away from the leaf to the stem
- B. They absorb sunlight that passes through the epidermis
- C. They store extra water for the leaf to use at midday
- D. They let carbon dioxide diffuse easily to the cells inside the leaf**

Correct answer: D. Air spaces let CO₂ that enters through the stomata diffuse to the mesophyll cells. Water is carried in xylem and sugars in phloem, and light is captured by chloroplasts.

USE AFTER: EXPLAIN PHASE — Explain: expert group for vascular bundles presents its function statement

Q4. A vascular bundle in a leaf contains xylem and phloem. Which statement describes their jobs correctly?

- A. Xylem brings water and minerals to the leaf and phloem carries sugars away**
- B. Xylem carries sugars out of the leaf and phloem brings water into the leaf
- C. Xylem and phloem both carry only carbon dioxide to the palisade cells
- D. Xylem lets in light and phloem controls the opening of stomata

Correct answer: A. Xylem supplies water and minerals to the leaf cells, and phloem exports the sugars made in the leaf. The distractors swap the two jobs or give them roles that belong to other structures.

USE AFTER: EXPLAIN PHASE — Explain: expert group for epidermis/cuticle presents its function statement

Q5. The cuticle and the epidermis of a leaf are transparent (clear). Why is this useful to the plant?

- A. It allows water vapour to escape freely from the whole leaf surface
- B. It lets air spaces fill with sunlight at night
- C. It helps the xylem to see where water is needed
- D. It lets light pass through to the chloroplasts in the cells below**

Correct answer: D. Transparent outer layers admit light so it can reach the chloroplasts in the palisade cells. A clear covering does not help water escape, and the other options make no biological sense.

USE AFTER: EXPLAIN PHASE — Explain: expert group for stomata/guard cells presents its function statement

Q6. What is the job of the two guard cells around a stoma?

- A. To protect the leaf by making a thick waxy layer
- B. To carry water from the roots to the leaf
- C. To open and close the stoma, controlling gas exchange and water vapour loss**
- D. To make sugars using sunlight

Correct answer: C. Guard cells change shape to open or close the stoma. This regulates how much CO₂ enters and how much water vapour leaves. Sugar-making happens in chloroplasts, and water is carried in xylem.

USE AFTER: EXPLAIN PHASE — Explain: synthesis question 'if midday sun makes stomata open wide, what does that mean for water loss?' and the 3-sentence written response

Q7. At midday in bright sun, a kale plant's stomata are wide open. The soil is still moist, yet the leaves wilt. Which explanation fits the leaf structure best?

- A. The guard cells close completely and trap heat inside the leaf
- B. Water vapour leaves through the open stomata faster than the xylem can replace it, so the cells lose turgor**
- C. The roots have permanently died, so no water can enter the plant
- D. Too much water enters the leaf cells and they burst

Correct answer: B. Open stomata let water vapour escape quickly. When loss is faster than xylem supply, the cells lose turgor and the leaf wilts. The plant recovers overnight, so the damage is not permanent.

USE AFTER: END OF LESSON — End of lesson: revisiting the sukuma wiki shade observation (pallor and slow growth)

Q8. A kale plant is moved under the shade of a mango tree. Wilting stops, but over several weeks the leaves turn pale and the plant grows slowly. Which explanation uses leaf structure correctly?

- A. The shade blocks the xylem, so no water can reach the leaf
- B. The shade makes the guard cells lose their pores
- C. Less light reaches the palisade cells, so the leaf makes less food and the plant grows slowly**
- D. The shade makes the cuticle so thick that air cannot pass through

Correct answer: C. Palisade cells hold most of the chloroplasts, and in shade they receive less light. This reduces the food the plant can make, causing pallor and slow growth. The other options describe changes that shade does not cause.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Lesson 1 sukuma wiki model with a leaf cross-section and arrows for light, water and water vapour

Q9. A student is revising her model of a kale leaf in a cross-section drawing. Which set of arrows is correct?

- A. Light enters through the stoma, water moves down the xylem, and water vapour leaves through the vascular bundle
- B. Light enters from above through the epidermis, water moves up the xylem, and water vapour leaves through a stoma**
- C. Light enters from below, water moves down the phloem, and water vapour enters through the cuticle
- D. Light enters from above, sugars move up the xylem, and water vapour leaves through the palisade cells

Correct answer: B. Light enters from above through the transparent cuticle and epidermis. Water travels up the xylem, and water vapour exits through the stomata. The other sets mix up directions or structures.

USE AFTER: DQB PHASE — DQB: green sticky note, a Lesson 1 question students can now partially answer

Q10. After observing a leaf cross-section, which Driving Question Board question could students already partly answer using today's evidence?

- A. Why do sukuma wiki leaves wilt at midday even when the soil is moist?**
- B. What exactly makes the guard cells open the stoma?
- C. Why are leaves green?
- D. What is the full chemical equation for photosynthesis?

Correct answer: A. Today's structures (open stomata, water vapour loss, xylem supply) give a partial answer to the wilting question. The other questions concern mechanisms and ideas the lesson has not yet investigated.